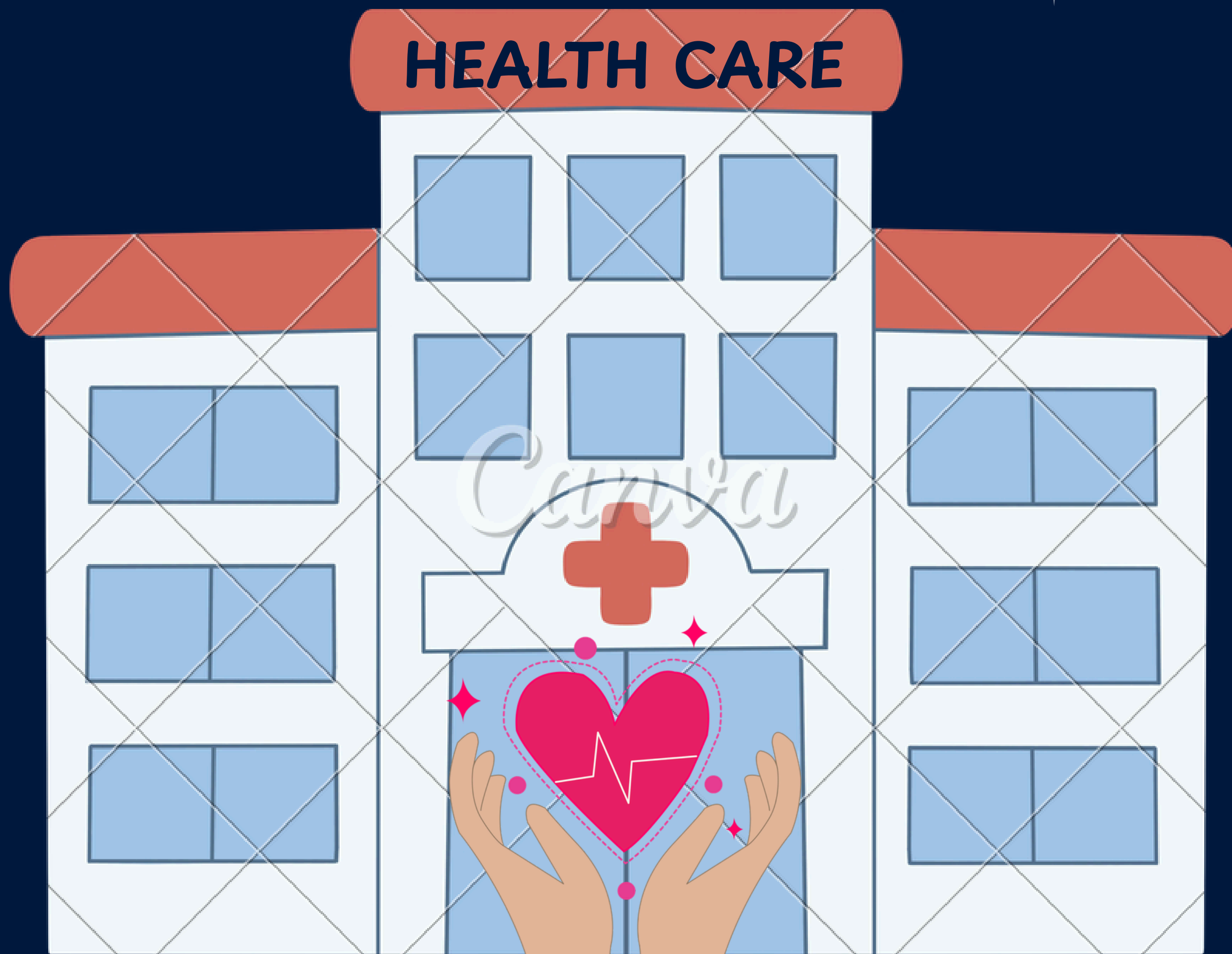


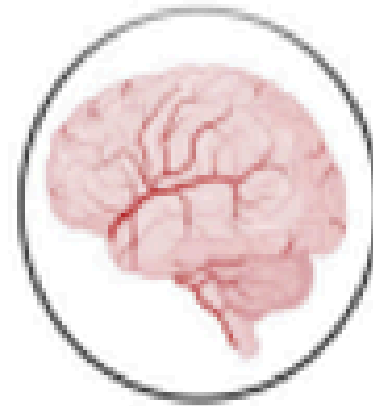


HEALTH CARE



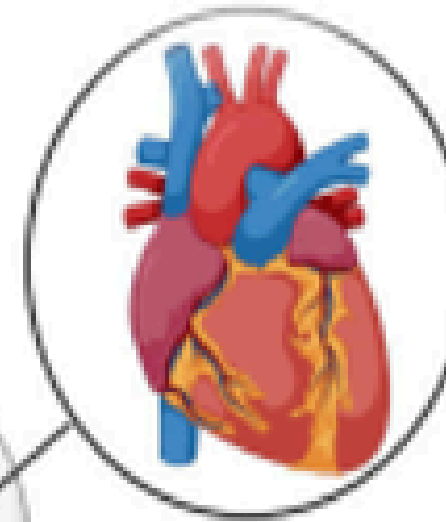
Neural System DT

- Brain and nervous system modeling and simulation
- Neurodegenerative disease modeling



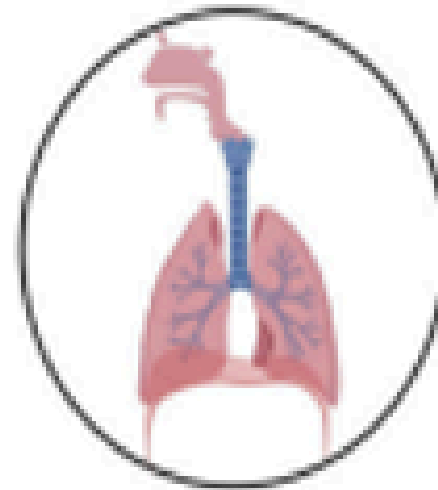
Cardiovascular System DT

- Circulation modeling and simulation
- Drug safety and arrhythmia assessment
- Hemodynamic modeling
- Heart failure and surgery



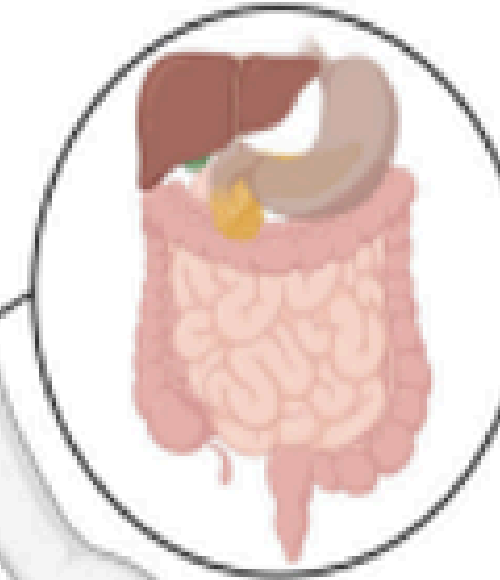
Respiratory System DT

- Lung modeling and simulation
- Respiratory monitoring



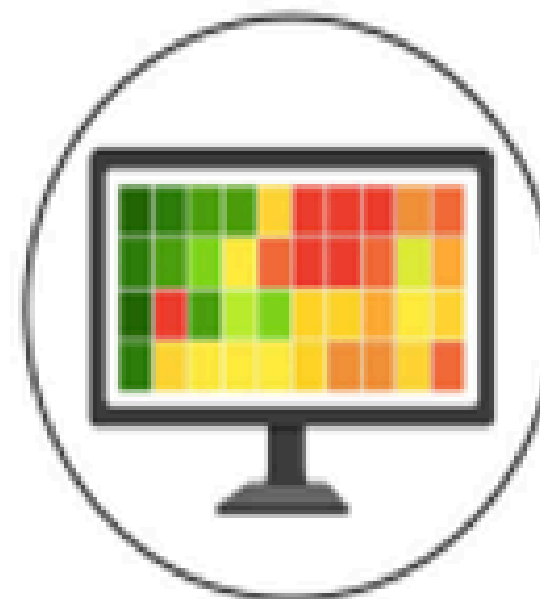
Metabolic and Endocrine System DT

- Diabetes Management
- Insulin resistance
- Metabolic response
- Hepatic system
- Pediatric metabolism



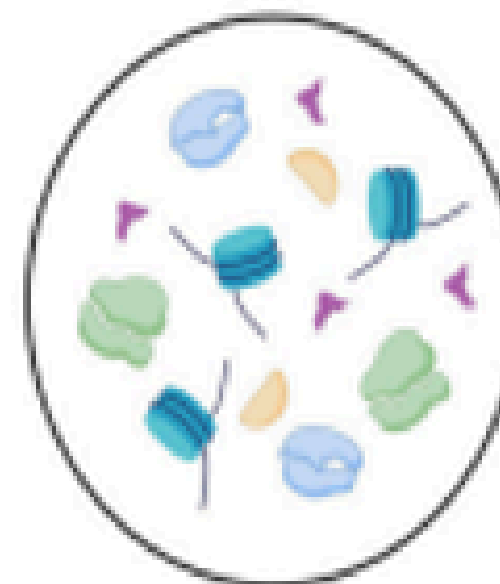
Cross-System DTs

- Surgical and interventional systems
- Treatment optimization
- Medical education
- Healthcare infrastructure
- Clinical operations
- Chronic condition management



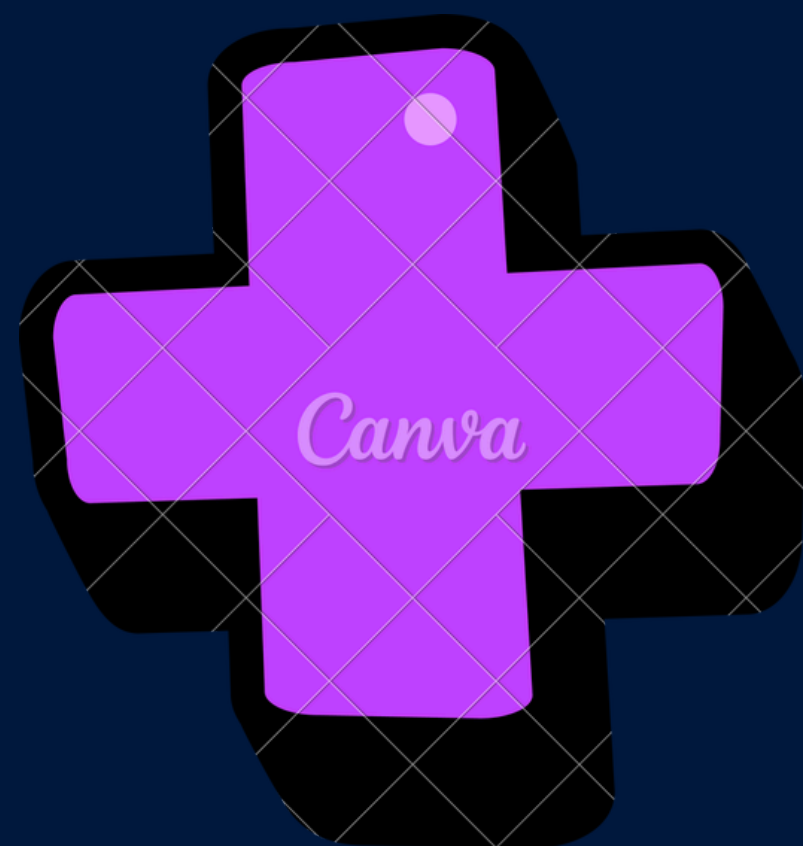
Cellular and Molecular DT

- Predicting short chained fatty acid production
- Drug development
- Molecular communication
- Genetic dynamics
- Perturbation prediction modeling

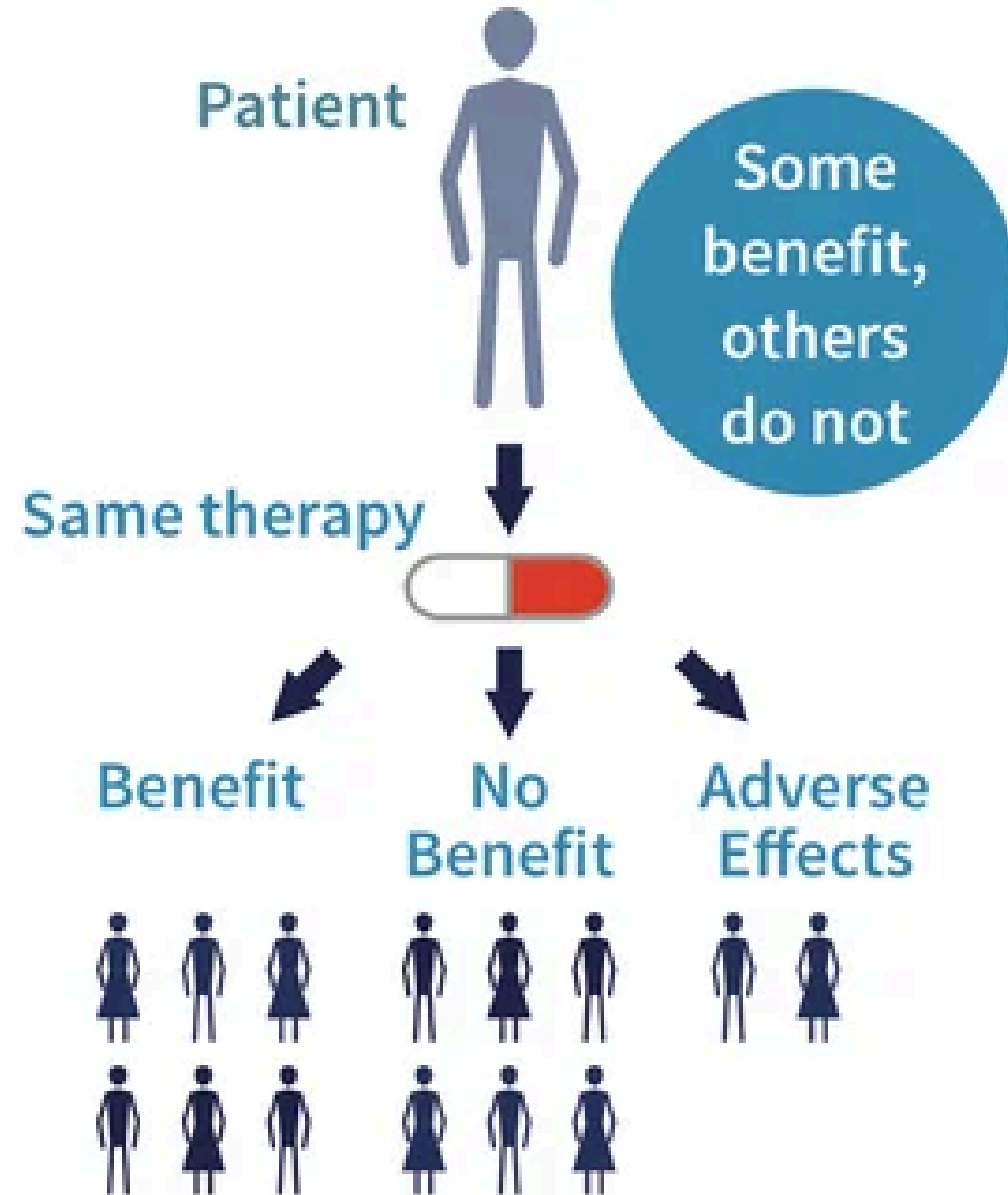


Why are we
using such
technology?!

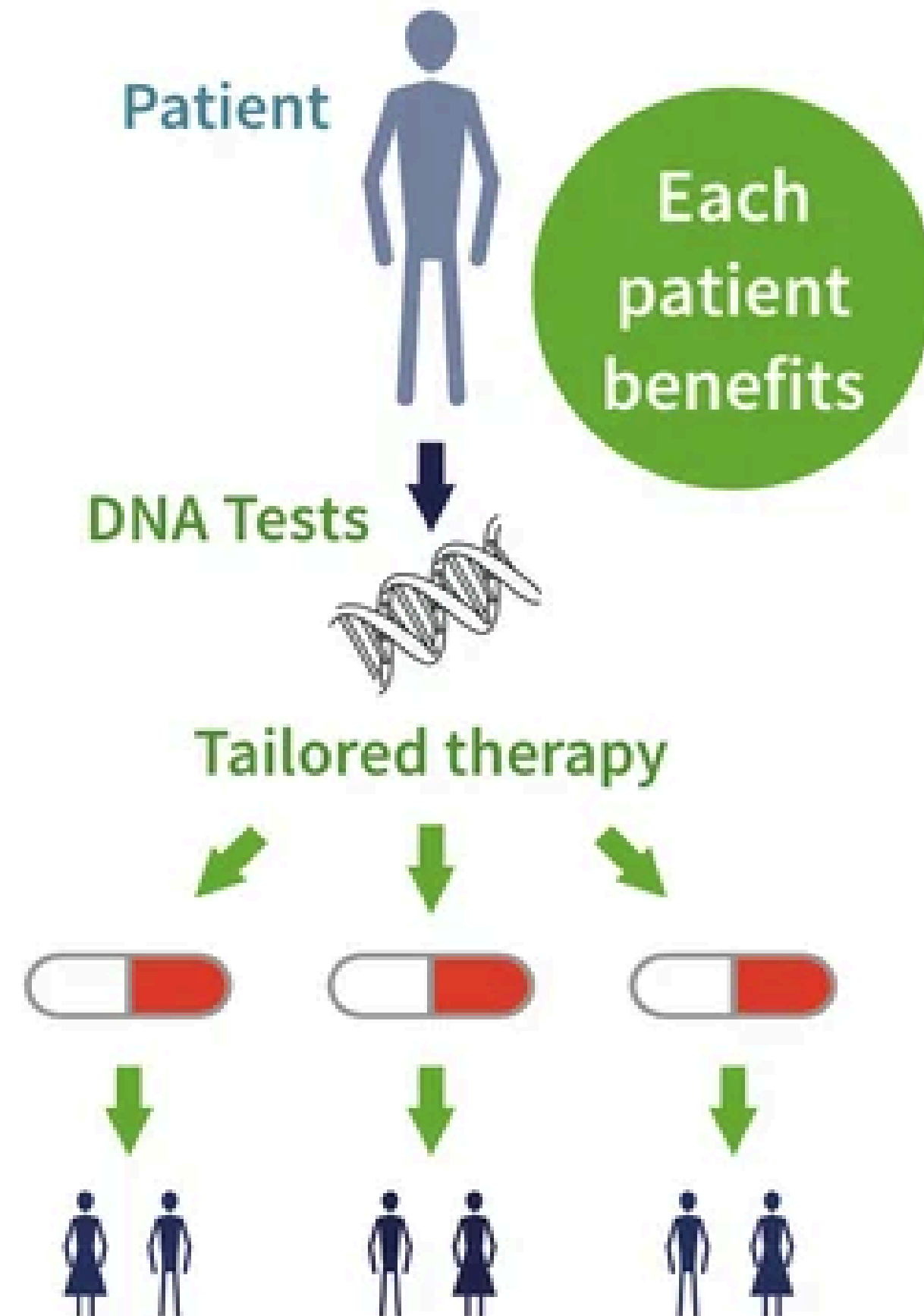




Without precision medicine



With precision medicine



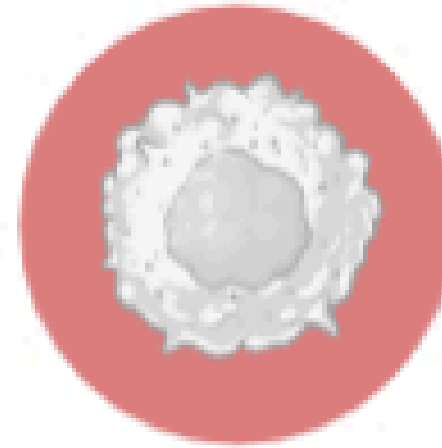
Clinical applications of DT in PM



Cardiology



Prediction of
response to CRT



Oncology



Tumor
simulation



Neurology



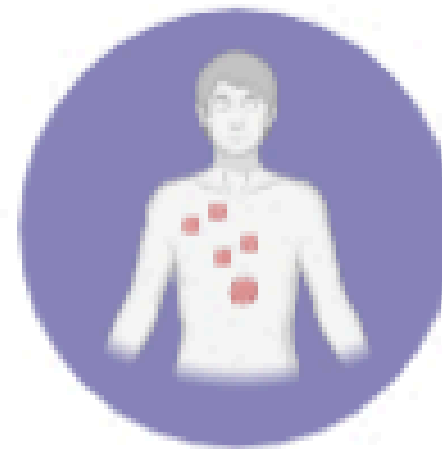
Planning of DBS



Pharmacogenomics



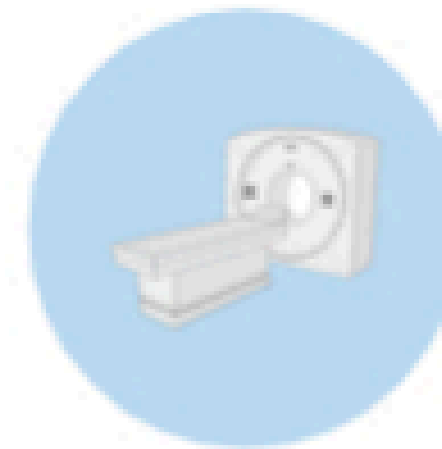
Dose
optimization



Rare Diseases



Gene therapy
modeling



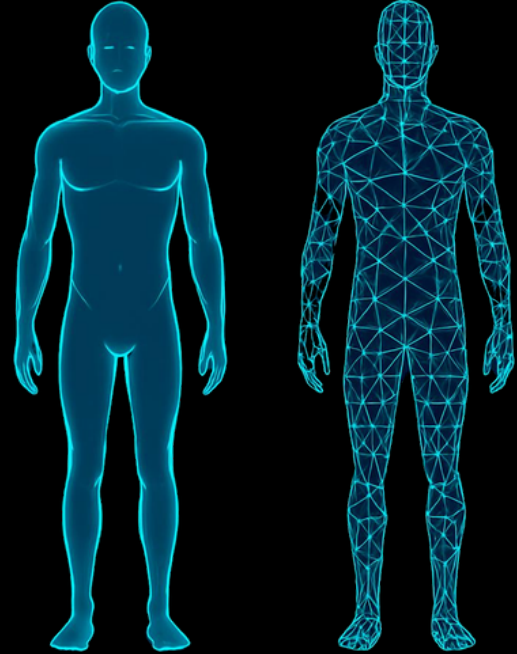
Medical Imaging

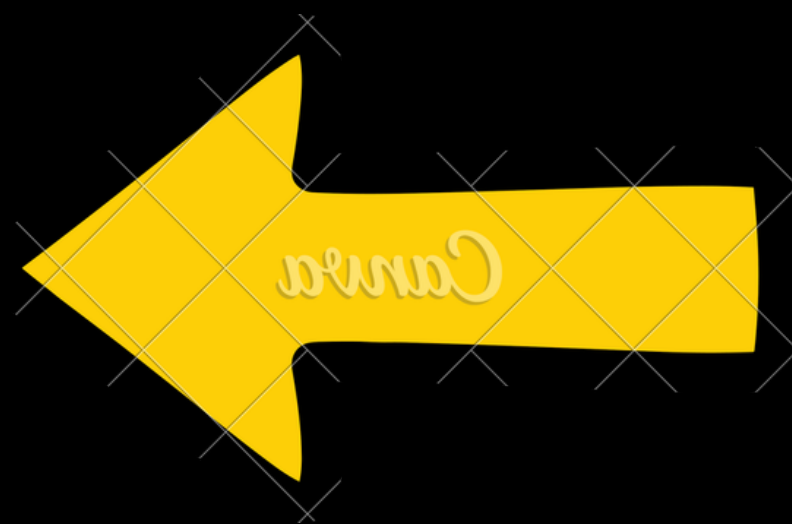
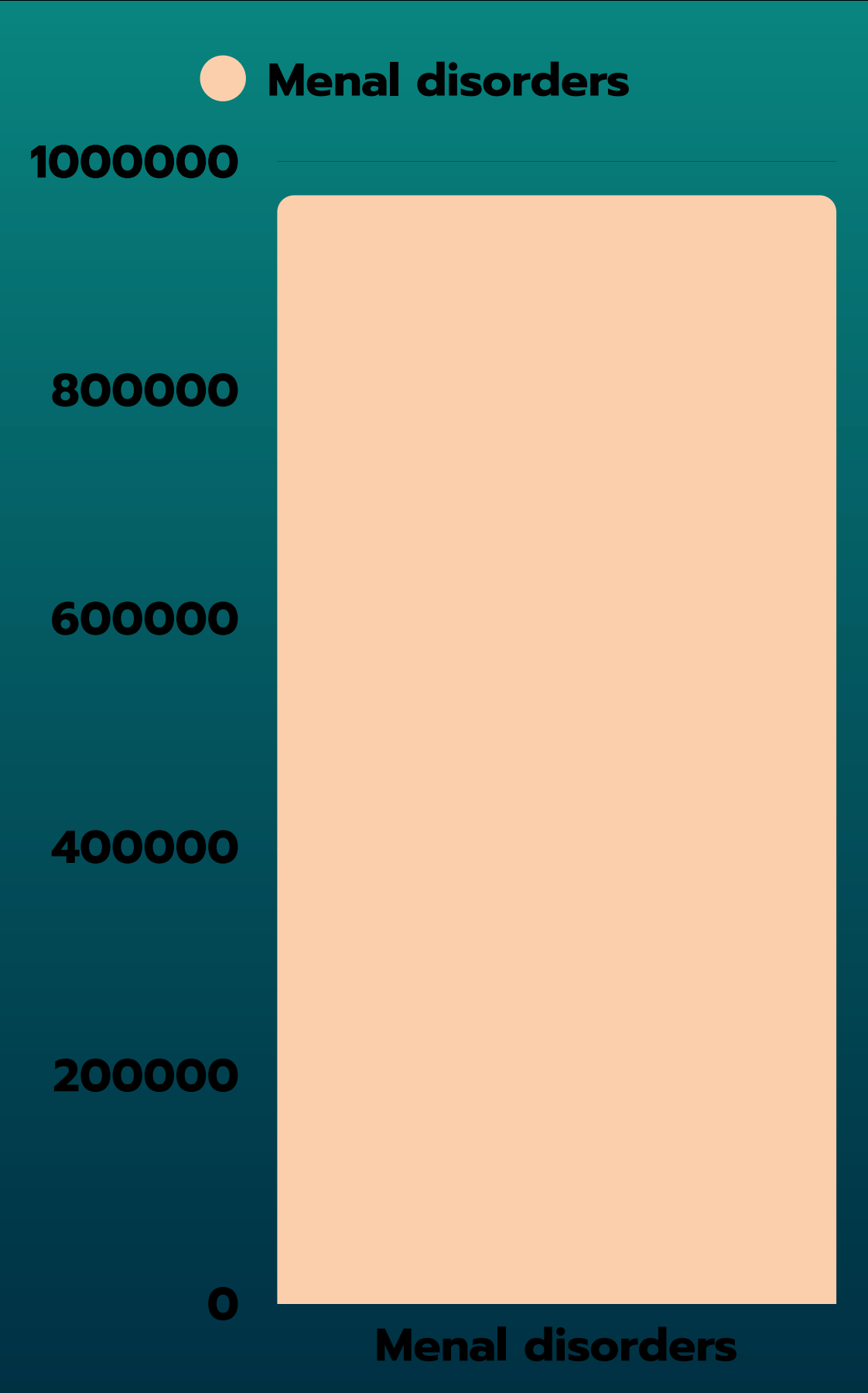


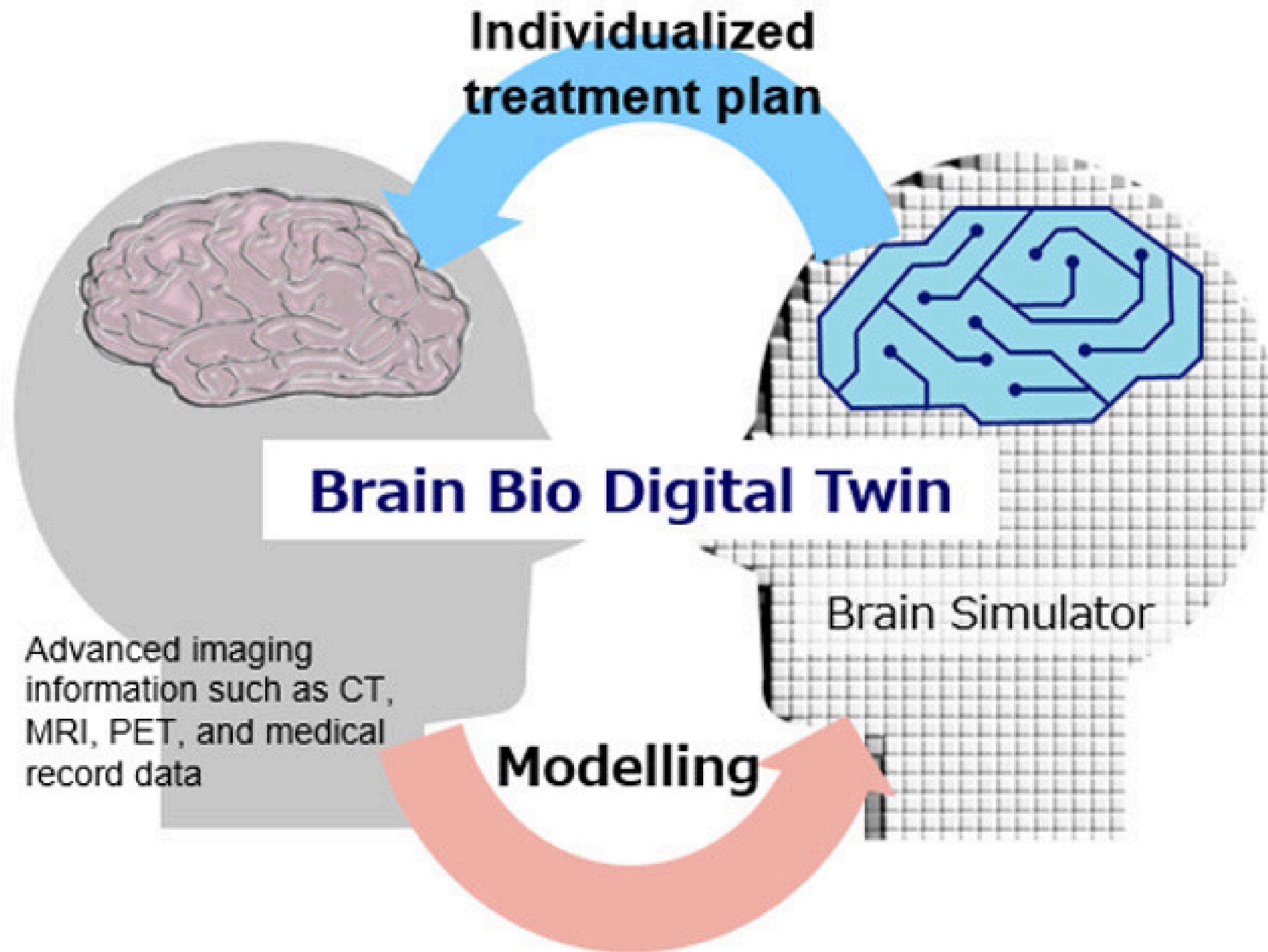
Estimation of
coronary flow

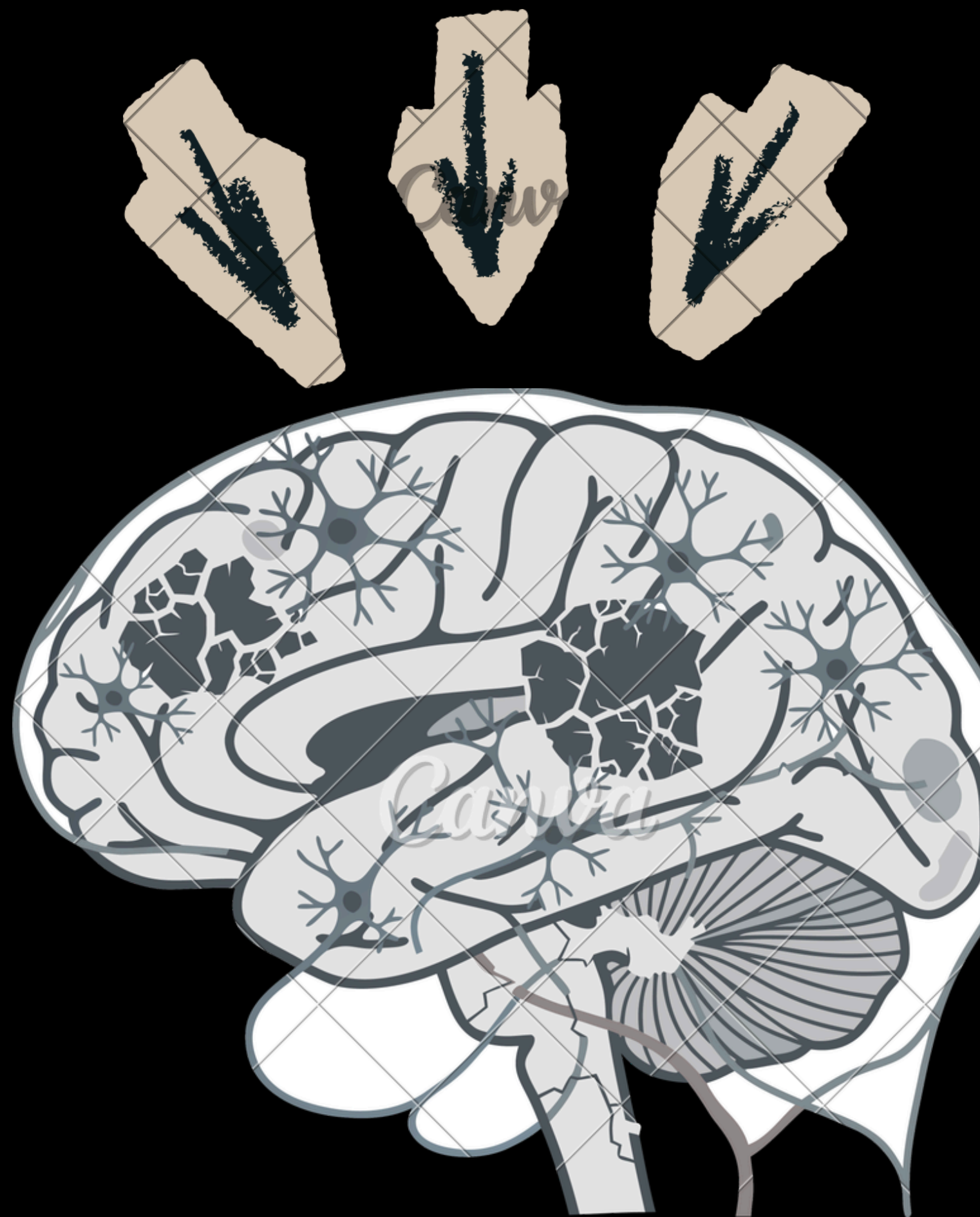


BRAIN & NEUROSCIENCE









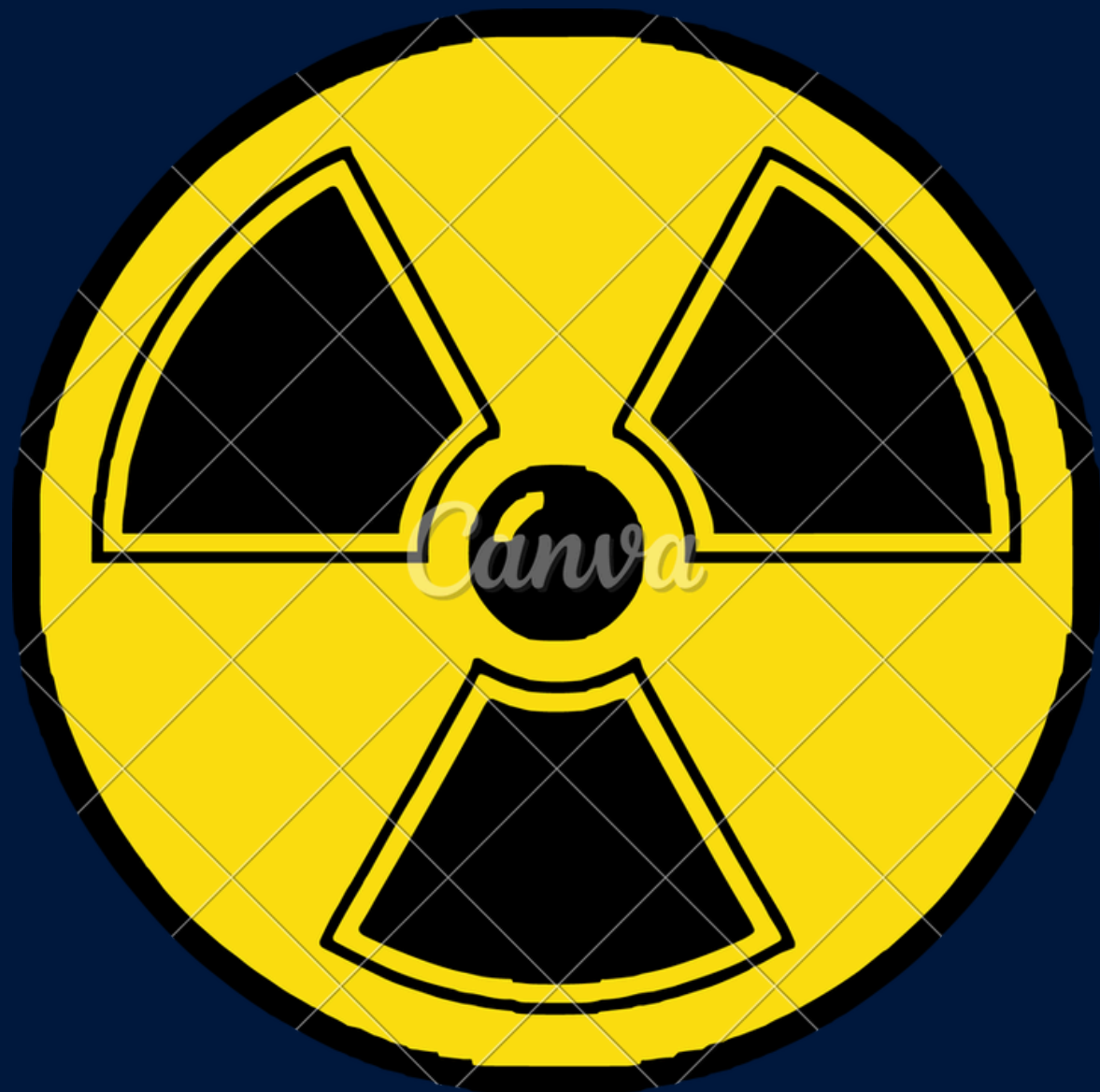
Neurodegenerative Disorders

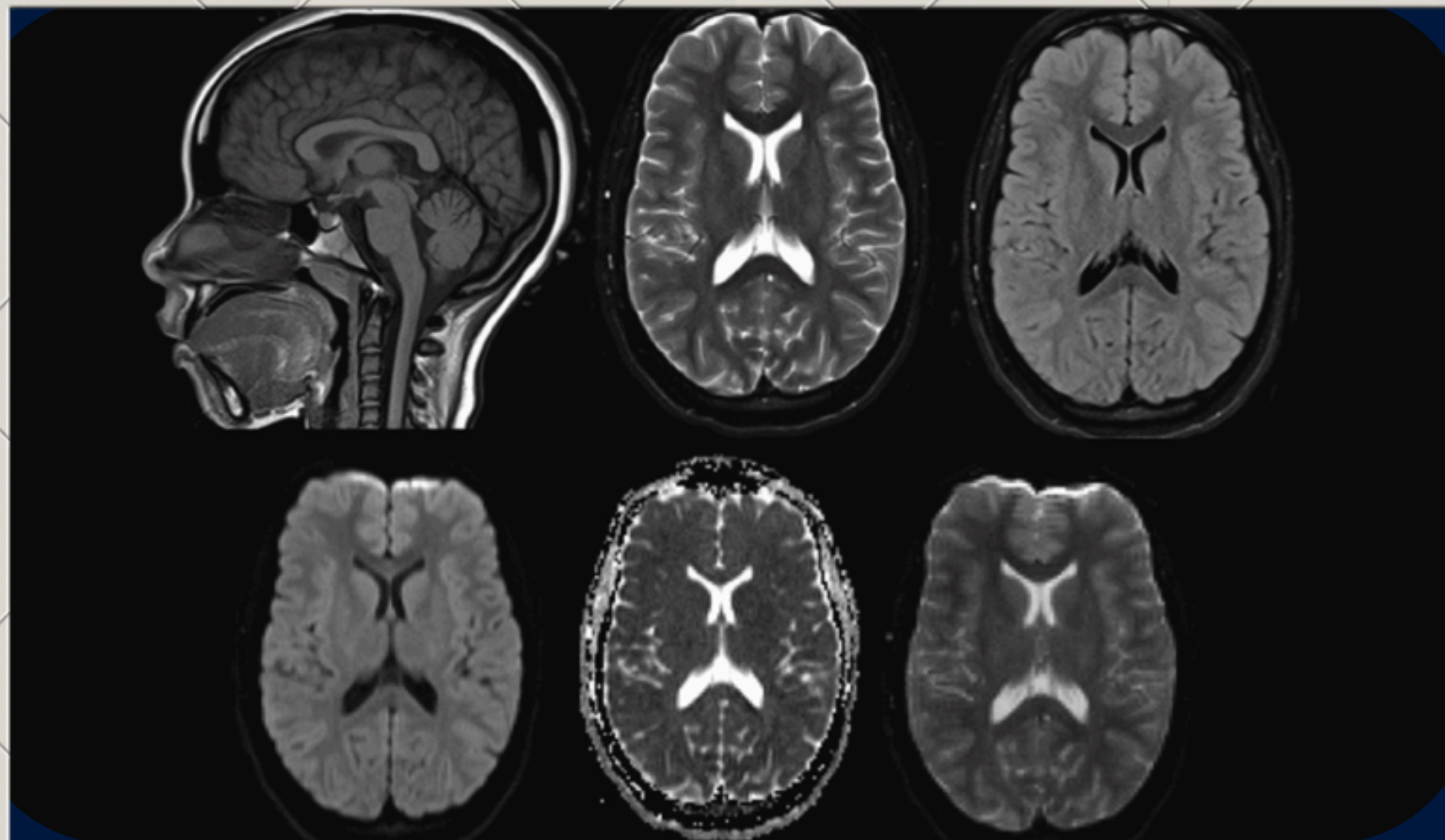
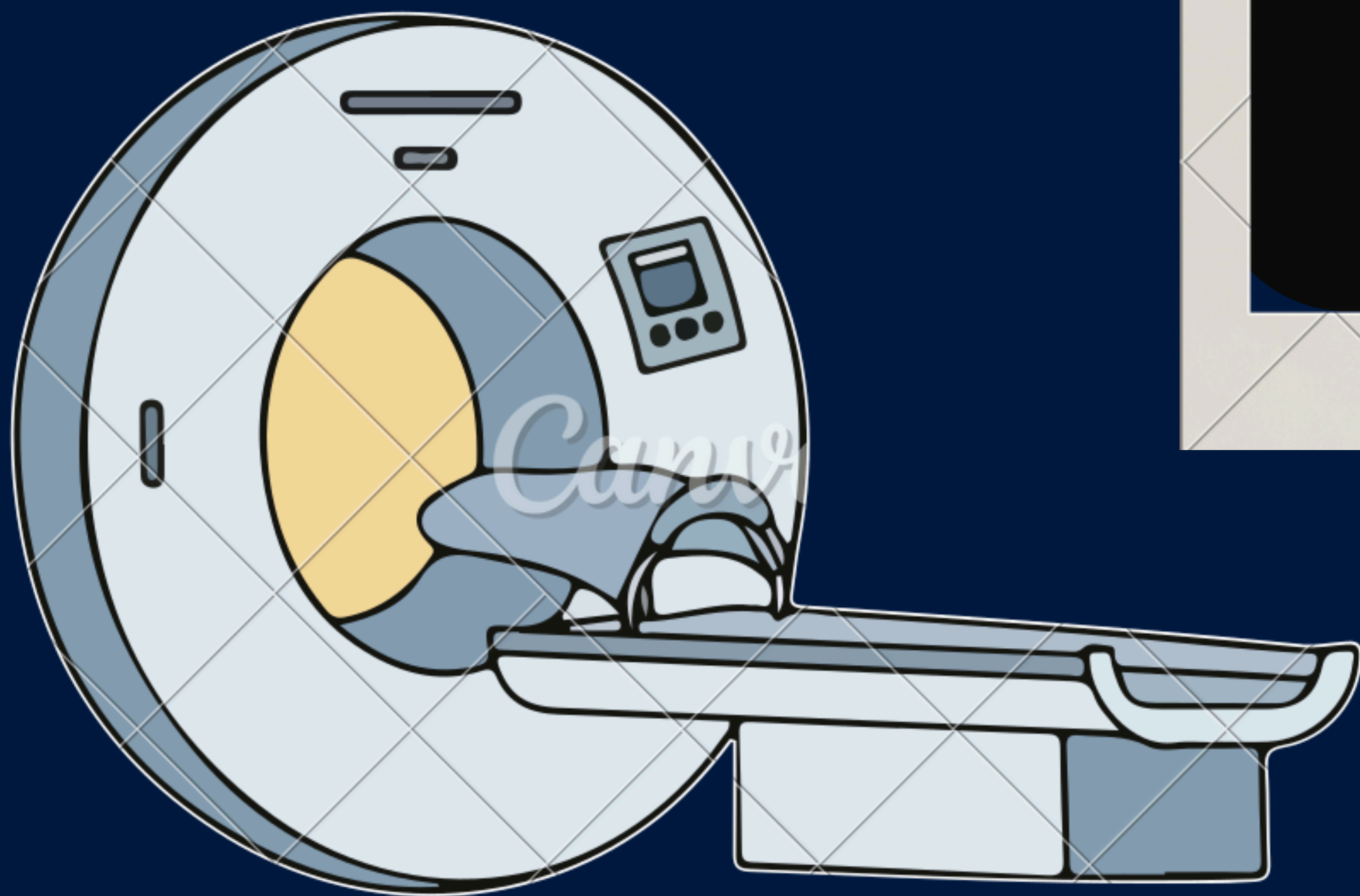
From Pathogenesis to
Clinical Approaches

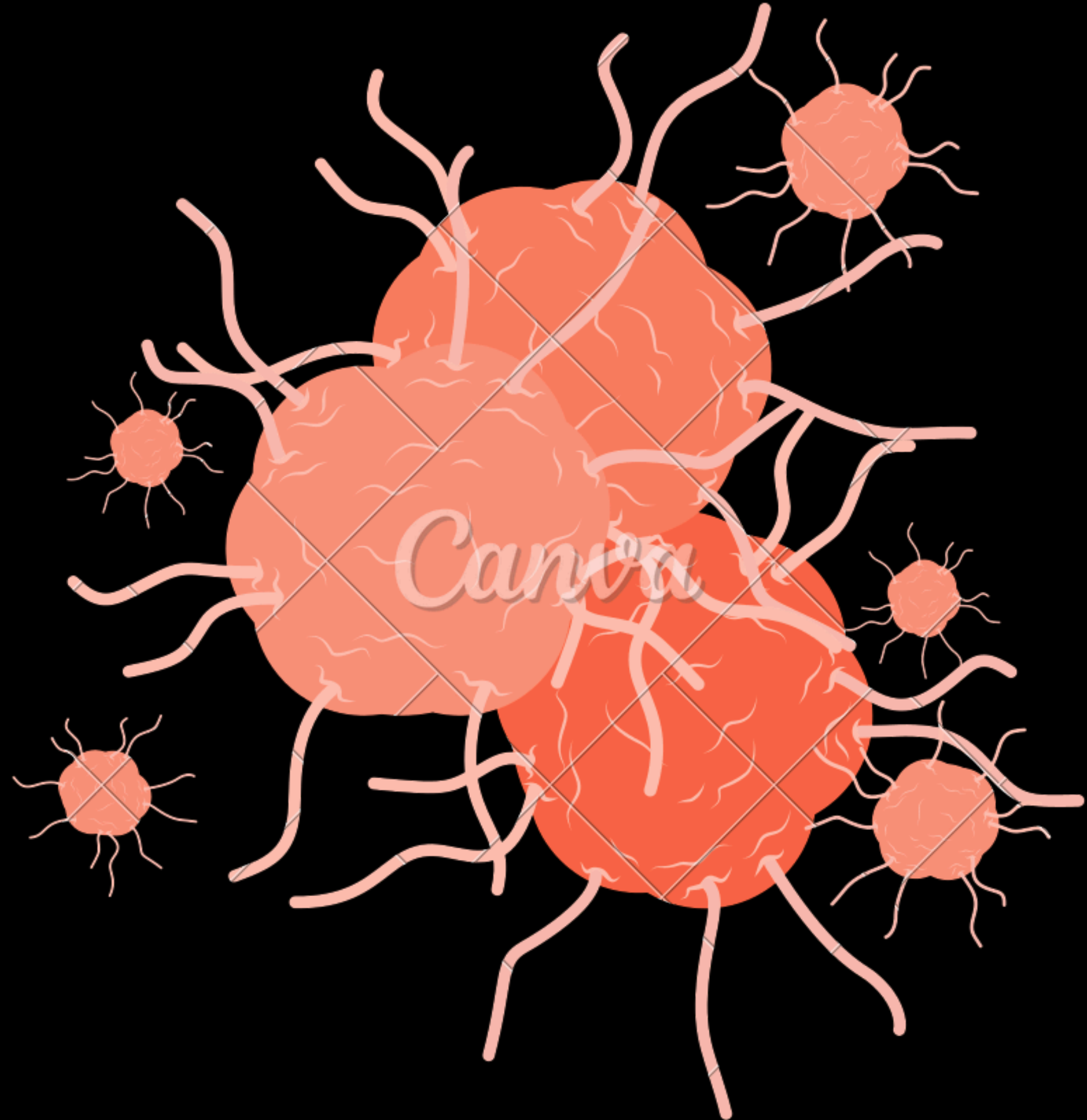
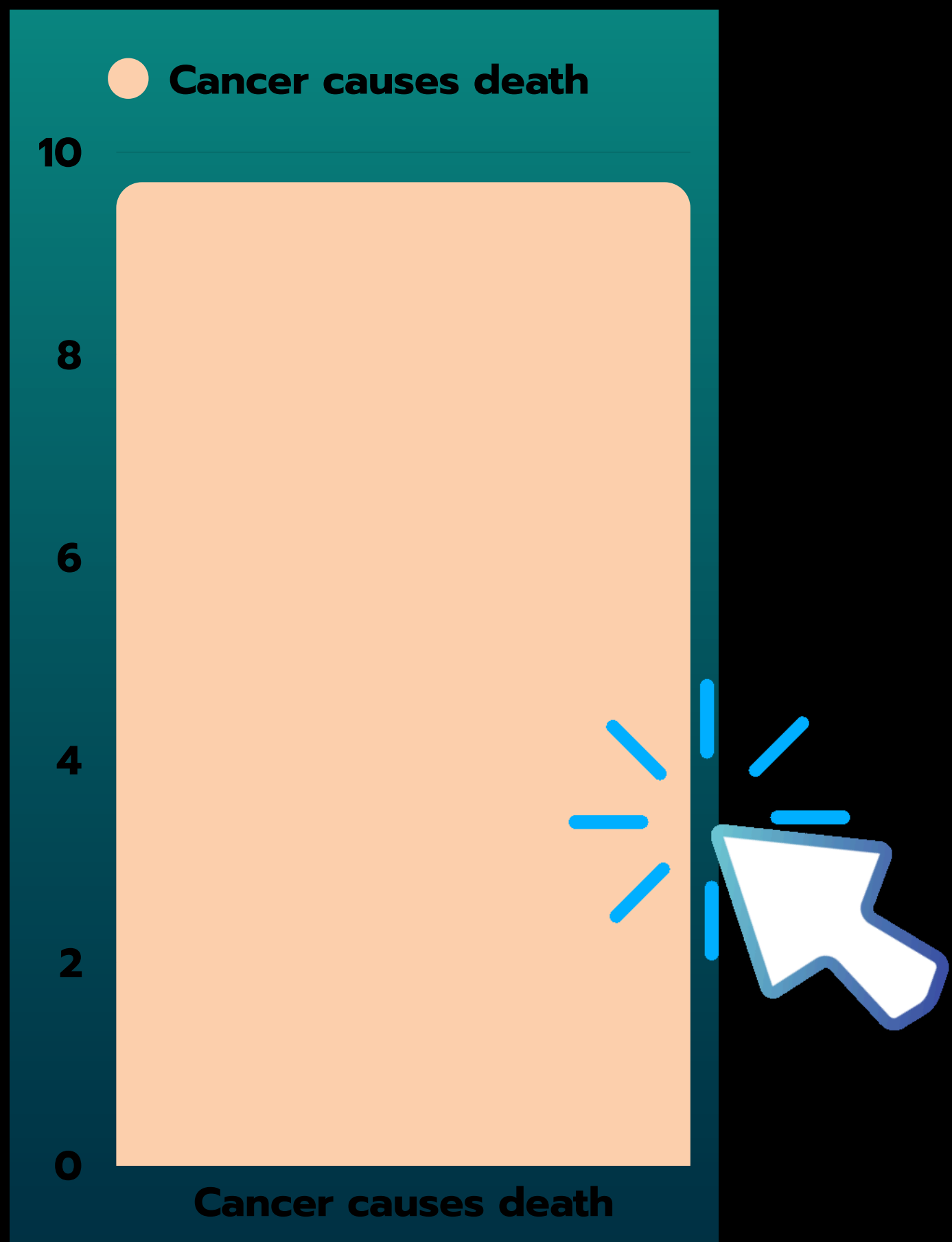


Mohit Agrawal • Manmohan Singhal
Amit Kumar Nayak • Shailendra Bhatt
Kamal Shah

**HOW THIS TECHNIQUE CAN HELP US
AS BIOPHYSICISTS?!!!!**









INTERNATIONAL WOMEN'S DAY



DIMENSIONS®

In 2022, about 2.3 million new cases of breast cancer were diagnosed globally, and there were about 665 000 deaths due to breast cancer.

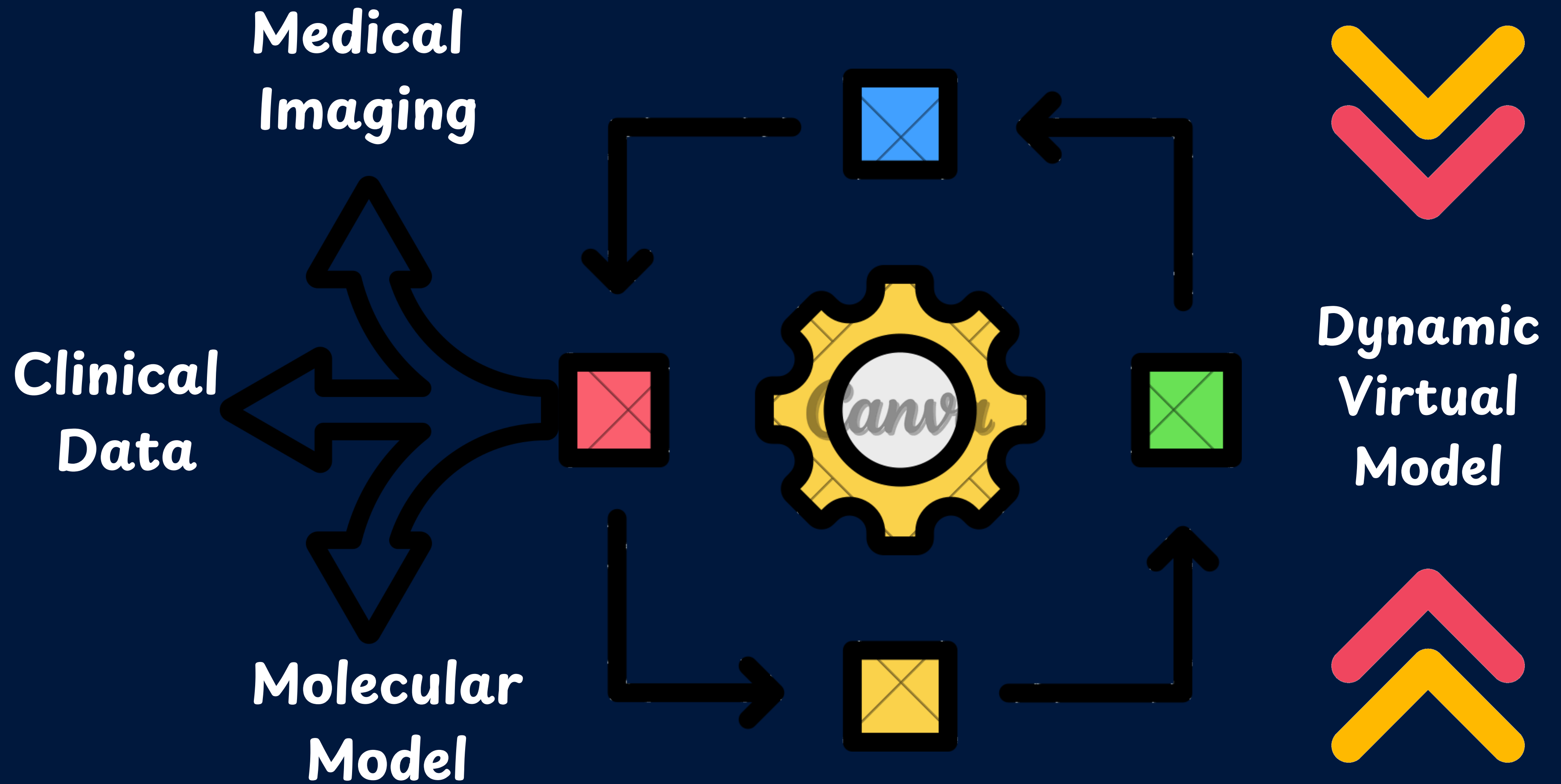


International Agency
for Research on Cancer

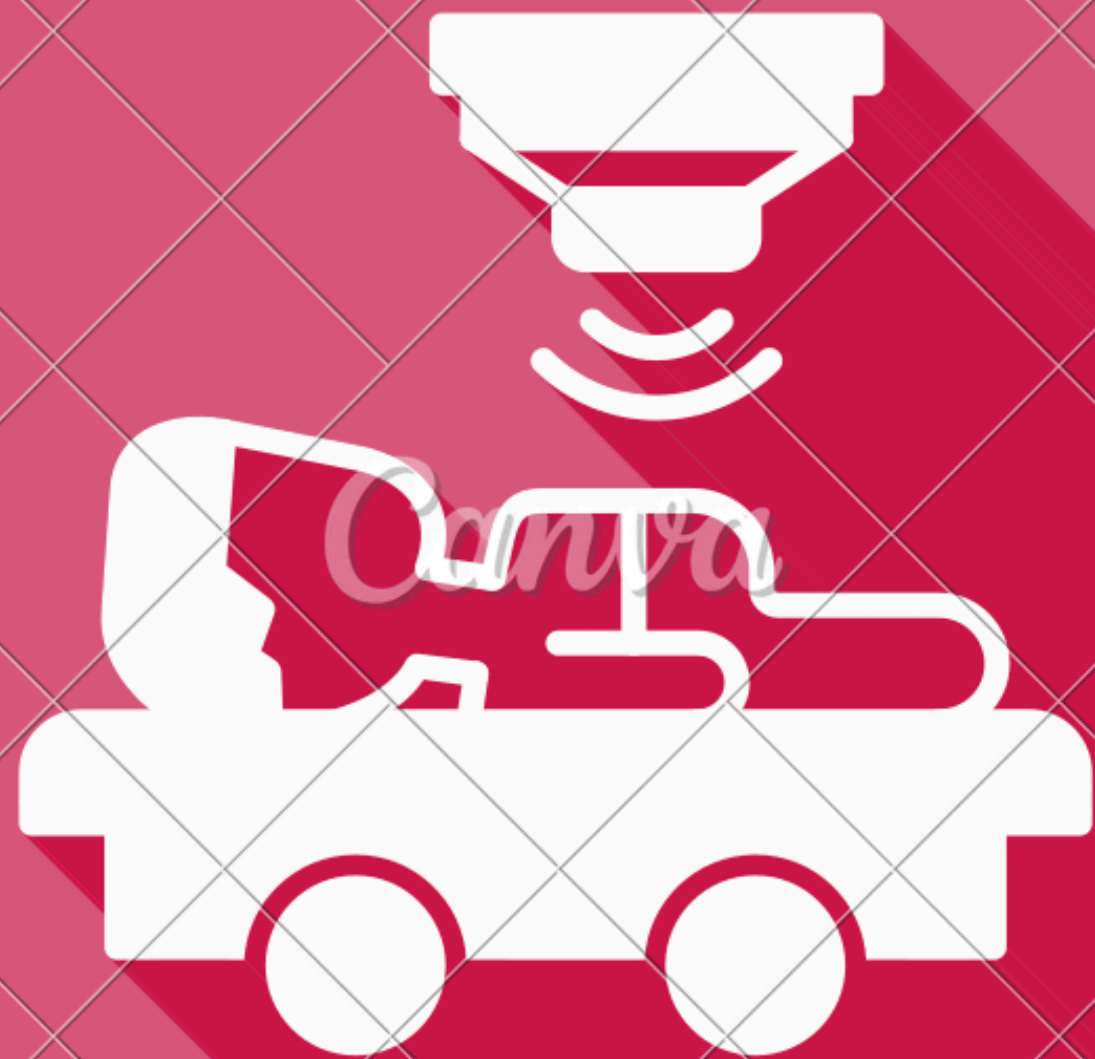


World Health
Organization

**BREAST CANCER
AWARENESS
MONTH 2024**



RADIOTHERAPY



CHEMOTHERAPY



UPDATE